Linear regression + Matching

ECON726

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Housekeeping

Paper selection assignment:

- Great work! Many of you have chosen excellent papers

Homework

- Read MHE 3.3 (Matching and Propensity Score)
  - Review 3.1. and 3.2., if you're still shaky on the concepts
- Read your paper. You won't understand everything; that's okay.
  - Maybe try to teach yourself a bit about the methodology. Read the relevant MHE chapter, or start reading about synthetic control
  - There is a language to papers. You're doing your best to understand it, even if you aren't fluent
- **Are you sure you want to do this paper? Does this seem like something you can do?**

Today

- Discuss paper selection
- Finish linear regression
- Introduction to matching
- Review examples in jupyter notebooks (regression; matching)

Review - Empirical Project Timeline

Paper selection: Due ~~**September 16**~~ **September 22** [5% of final grade]

- Done!

Project plan: Due **October 14** [15% of final grade]

- This paper should be your first effort at addressing points (1) and (2) above. Summarize the public policy or business context of the paper. Write down the specific question the paper is evaluating. Discuss the data your are using (both your replication data, and data for your extension) Discuss the extensions you are considering, and, if necessary, where you will be getting that data. **New: I'll ask you to verify that you've read the paper**

Replication results: Due **November 4**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 2** & **December 9** [10% of final grade]

- Prepare a 5-10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 16** [35% of final grade]

- Your paper should contain 5-8 pages of text, and 5-8 pages of figures (double spaced, 12 point font). You are expected to use python, and must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Paper selection

Thoughts on the selection process

- How confident are you in your paper?
- Were you able to find a topic you were excited in?
- Is there a causal question you'd rather look at, but you couldn't find a paper for?

Outline

Regression and causality

Regression gives the best linear approximation to the CEF, in a MMSE sense

- But when does a regression represent something causal?

A regression is **causal** when the CEF it approximates is causal

- Helpful to think of a CEF being causal by using the potential outcomes framework
- CEF is causal when it describes differences in average potential outcomes for a fixed reference population

Naïve comparison

Observed difference in earnings equals the average treatment effect (ATE) on the treated plus selection bias:

$$\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0] = \underbrace{\mathbf{E}[Y_1 - Y_0|D=1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y_0|D=1] - \mathbf{E}[Y_0|D=0]}_{\text{Selection bias}}$$

Random assignment ($Y_0, Y_1 \perp D$) will solve the selection problem

- Causes selection bias to equal zero

$$\mathbf{E}[Y_1|D=1] = \mathbf{E}[Y_1|D=0] = \mathbf{E}[Y_1] \quad \text{and} \quad \mathbf{E}[Y_0|D=0] = \mathbf{E}[Y_0|D=1] = \mathbf{E}[Y_0]$$

$$\implies \mathbf{E}[Y_0|D=1] - \mathbf{E}[Y_0|D=0] = 0$$

$$\implies \text{Selection bias is } 0$$

$$\implies \text{Naïve comparison = ATE on treated}$$

What do we do when we don't have random assignment?

Conditional Independence Assumption

Independence of potential outcomes and the treatment:

$\Longrightarrow$ $(Y_0, Y_1 \perp D)$

$\Longrightarrow$ randomization

What happens when we don't have independence? Meaning the occurrence of D influences the occurrence of Y_0 and Y_1 (or vice versa)?

Perhaps we can condition on certain variables X, such that the treatment D_i and the potential outcomes Y_{1i}, Y_{0i} LOOK independent

$$\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$$

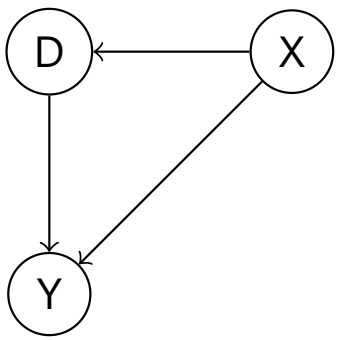
Conditional independence

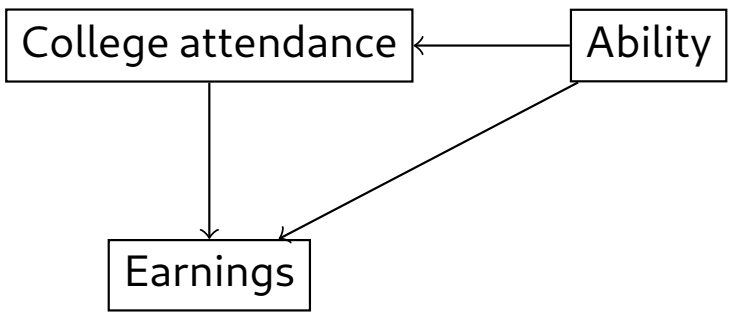
Conditional independence assumption:

$$\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$$

Intuitively: occurrence of D_i does not influence the occurrence of Y_{0i} or Y_{1i}, or vice versa, conditional on knowing X_i:

$$\mathbf{E}[Y_{0i}|D_i, X_i] = \mathbf{E}[Y_{0i}|X_i]$$
$$\mathbf{E}[Y_{1i}|D_i, X_i] = \mathbf{E}[Y_{1i}|X_i]$$





Selection bias under CIA

Under conditional independence assumption, selection bias disappears conditional on X_i:

$$\begin{aligned}\mathbf{E}[Y_{0i}|X_i, D_i = 1] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] =& \mathbf{E}[Y_{0i}|X_i, D_i = 0] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] \\ =& 0\end{aligned}$$

So we can write the difference in observed outcomes **conditional on** X_i as:

$$\begin{aligned}\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] &= \mathbf{E}[Y_{1i}|X_i, D_i = 1] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] \\ &= \mathbf{E}[Y_{1i}|X_i] - \mathbf{E}[Y_{0i}|X_i] \\ &= \mathbf{E}[Y_{1i} - Y_{0i}|X_i]\end{aligned}$$

Given the CIA, comparisons of average earnings (conditional on X_i) have a causal interpretation

When treatment takes on more than two values

Expanding CIA assumption to relations that can take on more than two values, such as when $s_i = \{0, 1, \ldots, 16\}$ equals years of schooling:

$$Y_{si} = f_i(s)$$

Individual specific function:

- Potential earnings that person i would receive after obtaining s years of education
- What would i learn for any value of schooling

If s could only take on two values, $s = 12$ and $s = 16$, then we're considering the value of college vs. no college:

$$Y_{0i} = f_i(12), \qquad Y_{1i} = f_i(16)$$

CIA with non-binary treatment

CIA in this more general set-up:

$$Y_{si} \perp s_i | X_i, \quad \forall s_i$$

- In randomized experiments (RCTs), s_i is randomly assigned based on covariates X_i
- In observational studies, the CIA means s_i is **as good as randomly assigned** conditional on X_i

Under conditional independence assumption, comparisons of average earnings across schooling levels have a causal interpretation:

$$\mathbf{E}[Y_i|X_i, s_i = s] - \mathbf{E}[Y_i|X_i, s_i = s-1] = \mathbf{E}[f_i(s) - f_i(s-1)|X_i]$$

For example, average causal effect of high school graduation:

$$\begin{aligned}\mathbf{E}[Y_i|X_i, s_i = 12] - \mathbf{E}[Y_i|X_i, s_i = 11] =& \mathbf{E}[f_i(12)|X_i, s_i = 12] - \mathbf{E}[f_i(11)|X_i, s_i = 11] \\ =& \mathbf{E}[f_i(12) - f_i(11)|X_i, s_i = 12]\end{aligned}$$

- Here, selection bias would come from the differences in potential earnings of high school graduates and non-graduates
- Under CIA, high school graduation is independent of potential earnings conditional on X_i, so selection bias disappears

CIA and regression

Regression is an empirical strategy that turns CIA into causal effects:

- Assume $f_i(s)$ is linear in s, and the same for everyone, except for an additive term η_i:

$$f_i(s) = \alpha + \rho s + \eta_i$$

- **Linear constant effects model**: equation is linear, and the functional relationship is the same for everyone
- Only individual-specific part of $f_i(s)$ is mean-zero η_i, which captures unobserved factors that determine potential earnings

Observed value will be:

$$Y_i = \alpha + \rho s_i + \eta_i,$$

Note that s_i is correlated with η_i: $\mathbf{E}[\eta_i | s_i] \neq 0$. If we ran this regression, we'd expect selection bias to impact our results.

CIA and regression (cont.)

Assume CIA holds, and η_i has the form:

$$\eta_i = X_i'\gamma + v_i$$

where:

- γ is a vector of population regression coefficients that satisfies $\mathbf{E}[\eta_i|X_i] = X_i'\gamma$
- v_i and X_i are uncorrelated by construction

Then, because of the CIA:

$$\begin{aligned}\mathbf{E}[f_i(s)|X_i, s_i] &= \mathbf{E}[f_i(s)|X_i] \\ &= \alpha + \rho s + \mathbf{E}[\eta_i|X] \\ &= \alpha + \rho s + X_i'\gamma\end{aligned}$$

The residual in the linear causal model v_i is uncorrelated with s_i and X_i, and ρ is the causal effect of interest:

$$Y_i = \alpha + \rho s_i + X_i'\gamma + v_i$$

$\Longrightarrow$ if we assume CIA holds in this linear causal model, ρ is a causal coefficient

Regression takeaways

We just discussed how CIA + linear causal model yields a causal interpretation of CEF

- Linear causal model: $Y_i = \alpha + \rho s_i + \eta_i$
- If CIA holds given some variables X_i: $Y_{0i}, Y_{1i} \perp s_i | X_i$
- Our regression function has a causal interpretation: $Y_i = \alpha + \rho s_i + X_i' \gamma + v_i$
  - ρ is the causal effect of s_i on Y_i
- CEF given by:

$$\mathbf{E}[Y_i | X_i, s_i] = \alpha + \rho s_i + X_i' \gamma$$

Thing to note:

- CIA is the same as unconfoundedness is the same as selection on observables
- In regression, CIA is the same as exogeneity

Motivating example

Consider the universe of applicants to the U.S. military:

- Y_i: average earnings
- D_i: dummy variable indicating whether i served in the military

We may be interested in the average effect of treatment on the treated (ToT):

$$\mathbf{E}[Y_{1i} - Y_{0i]}|D_i = 1]$$

Difference between:

- Average earnings of soldiers (observable)
- Average earnings of soldiers, had they not served (unobservable)

Naïve comparisons will be biased:

$$\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0] = \underbrace{\mathbf{E}[Y_{1i} - Y_{0i}|D_i = 1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y_{0i}|D_i = 1] - \mathbf{E}[Y_{0i}|D_i = 0]}_{\text{Selection bias}}$$

Motivating example: CIA

Conditional independence assumption:

$$Y_{0i}, Y_{1i} \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

$$\delta_{ToT} = \mathbf{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

Motivating example: CIA

Conditional independence assumption:

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Assuming CIA, selection bias disappears after conditioning on X_i:

$$\begin{aligned} \delta_{ToT} &= \mathbf{E}[Y_{1i} - Y_{0i} | D_i = 1] \\ &= \mathbf{E}\left[\mathbf{E}[Y_{1i} - Y_{0i} | X_i, D_i = 1] \middle| D_i = 1\right] && \because \text{LIE} \end{aligned}$$

Motivating example: CIA

Conditional independence assumption:

$$Y_{0i}, Y_{1i} \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

$$\begin{aligned}
\delta_{ToT} &= \mathbf{E}[Y_{1i} - Y_{0i} | D_i = 1] \\
&= \mathbf{E}\left[\mathbf{E}[Y_{1i} - Y_{0i} | X_i, D_i = 1] \middle| D_i = 1\right] && \because \text{LIE} \\
&= \mathbf{E}\left[\mathbf{E}[Y_{1i} | X_i, D_i = 1] - \mathbf{E}[Y_{0i} | X_i, D_i = 1] \middle| D_i = 1\right]
\end{aligned}$$

Motivating example: CIA

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\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_{0i} | X_i, D_i = 0] = \mathbf{E}[Y_{0i} | X_i, D_i = 1]$$

Motivating example: CIA

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&= \mathbf{E}\left[\mathbf{E}[Y_{1i} | X_i, D_i = 1] - \mathbf{E}[Y_{0i} | X_i, D_i = 0] \middle| D_i = 1\right]
\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_{0i} | X_i, D_i = 0] = \mathbf{E}[Y_{0i} | X_i, D_i = 1]$$

By definition of $Y_i = Y_{0i} + D_i(Y_{1i} - Y_{0i})$:

$$\begin{aligned}
\mathbf{E}[Y_i | X_i, D_i = 0] &= \mathbf{E}[Y_{0i} + D_i(Y_{1i} - Y_{0i}) | X_i, D_i = 0] \\
&= \mathbf{E}[Y_{0i} | X_i, D_i = 0] \\
\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_{1i} | X_i, D_i = 1]
\end{aligned}$$

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Conditional independence assumption:

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&= \mathbf{E}\left[\mathbf{E}[Y_{1i} | X_i, D_i = 1] - \mathbf{E}[Y_{0i} | X_i, D_i = 1] \middle| D_i = 1\right] \\
&= \mathbf{E}\left[\mathbf{E}[Y_{1i} | X_i, D_i = 1] - \mathbf{E}[Y_{0i} | X_i, D_i = 0] \middle| D_i = 1\right] \\
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$$\mathbf{E}[Y_{0i} | X_i, D_i = 0] = \mathbf{E}[Y_{0i} | X_i, D_i = 1]$$

By definition of $Y_i = Y_{0i} + D_i(Y_{1i} - Y_{0i})$:

$$\begin{aligned}
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&= \mathbf{E}[Y_{0i} | X_i, D_i = 0] \\
\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_{1i} | X_i, D_i = 1]
\end{aligned}$$

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1]} - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1\right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \underbrace{\mathbf{E}[Y_i|X_i, D_i = 0]}_{} \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

- $\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i
- $\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}\middle| D_i = 1\right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

$\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

Define δ_X to be the average difference in oberved Y_i for treated and untreated with a given value X_i:

$$\delta_X \equiv \underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}$$

- Example: difference in mean earnings by veteran status at each value of X_i

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1\right]$$

$$= \mathbf{E}\left[\delta_X \,\middle|\, D_i = 1\right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

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- Example: difference in mean earnings by veteran status at each value of X_i

Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Let's say there was some variable X_i that represents an indicator of whether an applicant is "high ability" or "lower ability"

- $X_i = 0$ means i is low ability
- $X_i = 1$ means i is high ability

We'll assume CIA: $\{Y_{0i}, Y_{1i}\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_{1i} - Y_{0i}|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0 | D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1 | D_i = 1)\end{aligned}$$

How do these elements connect to data?

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0 | D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1 | D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$ Difference in observed Y_i for treated and untreated with $X_i = 0$

(Average difference in wages by veteran status for **low** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$ Difference in observed Y_i for treated and untreated with $X_i = 0$
(Average difference in wages by veteran status for **low** ability i)

$\delta_{X_i=1}$ Difference in observed Y_i for treated and untreated with $X_i = 1$
(Average difference in wages by veteran status for **high** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0 | D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1 | D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
$\delta_{X_i=1}$	Difference in observed Y_i for treated and untreated with $X_i = 1$ (Average difference in wages by veteran status for **high** ability i)
$\mathbf{P}(X_i = 0 \mid D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
$\delta_{X_i=1}$	Difference in observed Y_i for treated and untreated with $X_i = 1$ (Average difference in wages by veteran status for **high** ability i)
$\mathbf{P}(X_i = 0\|D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)
$\mathbf{P}(X_i = 1\|D_i = 1)$	Probability $X_i = 1$ conditional on being treated (Fraction of veterans that are **high** ability)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
$\delta_{X_i=1}$	Difference in observed Y_i for treated and untreated with $X_i = 1$ (Average difference in wages by veteran status for **high** ability i)
$\mathbf{P}(X_i = 0\|D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)
$\mathbf{P}(X_i = 1\|D_i = 1)$	Probability $X_i = 1$ conditional on being treated (Fraction of veterans that are **high** ability)

Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect

Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Now instead of X_i being a dummy, assume X_i is the result of test of ability/intelligence

- $X_i \in \{0, 1, \ldots, 100\}$ and measures ability

We'll assume CIA: $\{Y_{0i}, Y_{1i}\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x=0}^{100} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1) + \cdots + \delta_{100} \mathbf{P}(X_i = 100|D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect

Now instead of X_i isn't a single variable, but instead a vector of (discrete) variables

- Year of birth, test score, year of application to the military, educational attainment at the time of application

Think of each realized $X_i = x$ as a particular combination of all these variables

- For example, $X_i = z$ might describe an applicant born in 1970, who scored a 95 on an ability test, and graduated from college

We'll assume CIA: $\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x | D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- The δ_x are x-specific treatment-control comparisons
  - For $X_i = z$, compare average wage difference of veterans and non-veterans who were born in 1970, who scored a 95 on an ability test, and graduated from college
  - Difference in average wages by veteran status for those with the same X_i
- Weighted together to produce a single overall treatment effect
  - Calculate weights based on the distribution of characteristics X_i among veterans

Unconditional Average Treatment Effect

Conditional treatment effect on treated:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_{1i} - Y_{0i}|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

- How much the typical soldier gained or lost as a result of enlisting

Can also write unconditional average treatment effect:

$$\begin{aligned}\delta_{ATE} &= \mathbf{E}\left[\mathbf{E}[Y_{1i}|X_i, D_i = 1] - \mathbf{E}[Y_{0i}|X_i, D_i = 0]\right] \\ &= \mathbf{E}[Y_{1i} - Y_{0i}] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x)\end{aligned}$$

- How much the typical applicant would gain or lose as a result of enlisting

Matching and regression

Matching and regression are related:

- Both are comparing outcomes (differences in Y_i) over treatment groups, controlling for covariates X_i
- These methods differ in implementation

Regression estimates of matching strategy:

$$Y_i = \sum_x d_{ix}\alpha_x + \delta_R D_i + e_i$$

where:

- $d_{ix} = \mathbf{1}[X_i = x]$: dummy variable that equals 1 if $X_i = x$, 0 otherwise
- α_x: regression effect for $X_i = x$
- δ_R: regression estimand

Matching and regression differences

In general, estimates from matching and from regression will often be similar, but not identical.

- Regression can be thought of as a type of matching strategy
  - Average effects are calculated by covariates X: δ_X
  - These effects are weighed to produce a weighted average
- Difference in how weights are determined. For ATE on treated:
  - Matching: weights are determined by distribution of covariates on treated
  - Regression: uses treatment-variance weighted average
  - These will differ unless treatment is independent of covariates

Note: neither regression nor covariate matching estimands give any weights to covariate cells that do not contain treated and control observations

- Often called **common support**